THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET A

Chapter 1: **Orienting Yourself: The Use of Coordinates** | Class 9 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(b)** Quadrant II	signs of (x, y) are $(-, +)$
2	**(a)** on the y-axis	$x = 0$, so the point lies on the y-axis
3	**(b)** 10 units	$\sqrt{(6-(0))^2 + (8-(0))^2} = \sqrt{36+64} = 10$
4	**(a)** 13 units	$\sqrt{(4-(-1))^2 + (11-(-1))^2} = \sqrt{25+144} = 13$
5	**(c)** $(2,\ 3)$	$\left(\frac{5+(-1)}{2}, \frac{9+(-3)}{2}\right)$
6	**(a)** $(3,\ -7)$	$B = (2x_M - x_A,\ 2y_M - y_A)$
7	**(d)** 17 units	$\sqrt{64+225} = 17$
8	**(b)** $(8,\ -1)$	reflection in the y-axis changes the sign of the x-coordinate
9	**(b)** $a = b$	the coordinates are equal in order only when $a = b$
10	**(d)** False; they are different because $(x, y) \neq (y, x)$ when $x \neq y$	order of coordinates matters

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	$AB = \sqrt{(5-(2))^2 + (7-(3))^2} = \sqrt{9+16} = \sqrt{25} = \mathbf{5}$ units	1+1
12	$AB = \sqrt{(4-(-1))^2 + (-10-(2))^2} = \sqrt{25+144} = \sqrt{169} = \mathbf{13}$ units	1+1
13	$AB = \sqrt{(12-(0))^2 + (0-(5))^2} = \sqrt{144+25} = \sqrt{169} = \mathbf{13}$ units	1+1
14	$AB = \sqrt{(5-(-3))^2 + (3-(-3))^2} = \sqrt{64+36} = \sqrt{100} = \mathbf{10}$ units	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	$AB = \sqrt{36+0} = 6$; $AC = \sqrt{9+16} = 5$; $BC = \sqrt{9+16} = 5$. Two sides ($AC = BC = 5$) are equal, so the triangle is isosceles.	1+1+1
16	$AB = \sqrt{36+0} = 6$; $AC = \sqrt{9+16} = 5$; $BC = \sqrt{9+16} = 5$. Two sides ($AC = BC = 5$) are equal, so the triangle is isosceles.	1+1+1
17	$AB = \sqrt{36+0} = 6$; $AC = \sqrt{9+16} = 5$; $BC = \sqrt{9+16} = 5$. Two sides ($AC = BC = 5$) are equal, so the triangle is isosceles.	1+1+1
	OR: $AB = \sqrt{36+0} = 6$; $AC = \sqrt{9+16} = 5$; $BC = \sqrt{9+16} = 5$. Two sides ($AC = BC = 5$) are equal, so the triangle is isosceles.	1+1+1
18	$AB = 4 = DC$, $BC = 3 = AD$ (opposite sides equal). Diagonals $AC = BD = \sqrt{16+9} = 5$ (equal). A parallelogram with equal diagonals is a rectangle.	1+1+1
	OR: $AB = 4 = DC$, $BC = 3 = AD$ (opposite sides equal). Diagonals $AC = BD = \sqrt{16+9} = 5$ (equal). A parallelogram with equal diagonals is a rectangle.	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) $10 \times 10 = 100$ intersections. (ii) Same E–W street; $6 - 2 = 4$ blocks apart ⇒ $4 \times 200 = 800$ m. (iii) Shifts: $4 \times 200 = 800$ m and $3 \times 200 = 600$ m ⇒ $\sqrt{800^2 + 600^2} = 1000$ m.	1+2+2
	OR: (i) $PQ = \sqrt{36+64} = 10$. (ii) $M = (1, -1)$. $MP = \sqrt{9+16} = 5$, $MQ = \sqrt{9+16} = 5$. Each $= \frac{1}{2}PQ$ ✓	2+3
20	$AB = \sqrt{9+25} = \sqrt{34}$, $BC = \sqrt{25+9} = \sqrt{34}$, $CD = \sqrt{9+25} = \sqrt{34}$, $DA = \sqrt{25+9} = \sqrt{34}$ (all sides equal). $AC = \sqrt{4+64} = \sqrt{68}$, $BD = \sqrt{64+4} = \sqrt{68}$ (diagonals equal). Rhombus with equal diagonals ⇒ square.	3+2
	OR: (i) $LG = \sqrt{9+16} = 5$, $CG = \sqrt{9+16} = 5$ ✓. (ii) $M = (5, 3)$. (iii) $GM = 4$ units $= 40$ m; it is the altitude to the base LC of the isosceles triangle LGC.	1+2+2

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET B

Chapter 1: **Orienting Yourself: The Use of Coordinates** | Class 9 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(c)** Quadrant IV	signs of (x, y) are $(+, -)$
2	**(a)** on the x-axis	$y = 0$, so the point lies on the x-axis
3	**(d)** 13 units	$\sqrt{(-7-(5))^2 + (6-(1))^2} = \sqrt{144+25} = 13$
4	**(c)** $(4,\ 7)$	$\left(\frac{2+(6)}{2}, \frac{4+(10)}{2}\right)$
5	**(c)** $(-2,\ 3)$	$\left(\frac{-6+(2)}{2}, \frac{-2+(8)}{2}\right)$
6	**(d)** 10 units	$\sqrt{36+64} = 10$
7	**(a)** $(-3,\ -4)$	reflection in the y-axis changes the sign of the x-coordinate
8	**(d)** $(0,\ 0)$	intersection of the axes
9	**(a)** y-axis (measured along the x-axis)	definition of abscissa
10	**(d)** $(1,\ 5)$ and $(5,\ 5)$	shift vertically by 3

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Midpoint $= \left(\frac{4+(10)}{2}, \frac{6+(-2)}{2}\right) = (\mathbf{7},\ \mathbf{2})$	1+1
12	Midpoint $= \left(\frac{-5+(1)}{2}, \frac{3+(9)}{2}\right) = (\mathbf{-2},\ \mathbf{6})$	1+1
13	Midpoint $= \left(\frac{7+(-3)}{2}, \frac{-4+(2)}{2}\right) = (\mathbf{2},\ \mathbf{-1})$	1+1
14	Midpoint $= \left(\frac{0+(-8)}{2}, \frac{0+(10)}{2}\right) = (\mathbf{-4},\ \mathbf{5})$	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	$AB = 5 = DC$, $BC = 3 = AD$ (opposite sides equal). Diagonals $AC = BD = \sqrt{25+9} = \sqrt{34}$ (equal). A parallelogram with equal diagonals is a rectangle.	1+1+1
16	$AB = 6 = DC$, $BC = 5 = AD$ (opposite sides equal). Diagonals $AC = BD = \sqrt{36+25} = \sqrt{61}$ (equal). A parallelogram with equal diagonals is a rectangle.	1+1+1
17	Let P be nearer A. $P = \left(\frac{2(2)+11}{3}, \frac{2(5)+14}{3}\right) = (5, 8)$; $Q = \left(\frac{2+2(11)}{3}, \frac{5+2(14)}{3}\right) = (8, 11)$	1+1+1
	OR: Let P be nearer A. $P = \left(\frac{2(-4)+8}{3}, \frac{2(1)+7}{3}\right) = (0, 3)$; $Q = \left(\frac{-4+2(8)}{3}, \frac{1+2(7)}{3}\right) = (4, 5)$	1+1+1
18	Sum of coordinates: if A, B, C are vertices with D mid of BC etc., then $A = E + F - D = (1, 7)$, $B = D + F - E = (-1, -1)$, $C = D + E - F = (11, 3)$. Check: midpoint of $BC = (5, 1) = D$ ✓	1+1+1
	OR: $MA = \sqrt{9+16} = 5$, $AG = \sqrt{36+64} = 10$, $MG = \sqrt{81+144} = 15$. $MA + AG = MG \Rightarrow$ collinear.	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	$AB = 5$, $BC = 5$, $AC = 10 \Rightarrow AB + BC = AC$ (collinear). B is the midpoint $\Rightarrow$ ratio $1 : 1$.	3+2
	OR: (i) $AB = 8$, $AC = BC = \sqrt{16+36} = 2\sqrt{13} \Rightarrow$ isosceles. (ii) $D(6, 3)$, $E(2, 3)$, $F(4, 0)$; $DE = 4$, $EF = \sqrt{4+9} = \sqrt{13}$, $FD = \sqrt{4+9} = \sqrt{13}$. Perimeter $= 4 + 2\sqrt{13}$ (half of that of ABC).	2+3
20	Angle at A is a right angle, so BC is a diameter. Centre $=$ midpoint of $BC = (3, 4)$; radius $= \frac{1}{2}\sqrt{36+64} = 5$.	3+2
	OR: (i) Rectangle. (ii) $AB = 4$, $BC = 6$; area $= 24$ sq. units; diagonal $= \sqrt{16+36} = 2\sqrt{13}$. (iii) Centre $= (-1, -1)$.	1+2+2

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.